

Linear#

<https://pytorch.org/docs/stable/generated/torch.nn.quantized.dynamic.Linear>

Description:

A dynamic quantized linear module with floating point tensor as inputs and outputs. We adopt the same interface as `torch.nn.Linear`, please see <https://pytorch.org/docs/stable/nn.html#torch.nn.Linear> for documentation. Similar to `torch.nn.Linear`, attributes will be randomly initialized at module creation time and will be overwritten later weight (Tensor) – the non-learnable quantized weights of the module which are of shape $(\text{out_features}, \text{in_features})$. bias (Tensor) – the non-learnable floating point bias of the module of shape (out_features) . If bias is True, the values are initialized to zero.

Function Signature

```
class torch.nn.quantized.dynamic.Linear(in_features,  
out_features, bias_=True, dtype=torch.qint8)[source]#
```

Variables

```
classmethod from_float(mod,  
use_precomputed_fake_quant=False)[source]#
```

Parameters

```
classmethod from_reference(ref_qlinear)[source]#
```

Code Examples

```
>>> m = nn.quantized.dynamic.Linear(20, 30)  
>>> input = torch.randn(128, 20)  
>>> output = m(input)  
>>> print(output.size())  
torch.Size([128, 30])
```

Detailed Documentation

Documentation

A dynamic quantized linear module with floating point tensor as inputs and outputs. We adopt the same interface as `torch.nn.Linear`, please see <https://pytorch.org/docs/stable/nn.html#torch.nn.Linear> for documentation.

Similar to `torch.nn.Linear`, attributes will be randomly initialized at module creation time and will be overwritten later

`weight` (Tensor) – the non-learnable quantized weights of the module which are of shape `(out_features, in_features)` (`\text{out\_features}`, `\text{in\_features}`) (`out_features, in_features`).

`bias` (Tensor) – the non-learnable floating point bias of the module of shape `(out_features)` (`\text{out\_features}`) (`out_features`). If `bias` is `True`, the values are initialized to zero.

Create a dynamic quantized module from a float module or `qparams_dict`

`mod` (Module) – a float module, either produced by `torch.ao.quantization` utilities or provided by the user

Create a (fbgemm/qnnpack) dynamic quantized module from a reference quantized module :param `ref_qlinear`: a reference quantized module, either produced by :type `ref_qlinear`: Module :param `torch.ao.quantization` functions or provided by the user:

